

ZHENXING GUO

Mailing address: School of Data Science (SDS)
Chinese University of Hong Kong, Shenzhen (CUHK-SZ)
320a Daoyuan Building
2001 Longxiang Road, Longgang District
Shenzhen, Guangdong, China

Email: guozhenxing@cuhk.edu.cn

Google scholar: <https://scholar.google.com.hk/citations?user=hm0mP4EAAAAJ&hl=zh-CN>

Employment

- **Nov. 2022 – Present, Assistant Professor (Tenure-track)** School of Data Science, Chinese University of Hong Kong, Shenzhen, Guangdong, China.

Education

- **Ph.D in Biostatistics** 2017 - 2022
Department of Biostatistics and Bioinformatics
Emory University, Atlanta, USA.
- **M.S. in Operational Research** 2014 - 2017
Academy of Mathematics and System Science (AMSS)
Chinese Academy of Sciences (CAS), Beijing, China.
- **B.S. in Applied Mathematics** 2010 - 2014
College of Mathematics & Econometrics
Hunan University, Changsha, China.

Previous Work Experience

- **June 2020 - August 2020, Statistical Technical Summer Intern** at Bristol Myers Squibb, Summit, NJ 07901, USA
Mentors: Rong Liu
Role: Augmentation of current clinical trial with external real world data using Bayesian techniques.

Research Interests

- Statistical Methodology: Epitranscriptomics and multi-omics data analyses. My detailed research interests lies in the development of statistically principled and computationally efficient method and tools for the analysis of biomedical data, in particular, omics data.

- Applications: Cancer, Alzheimer’s disease, asthma, cardiovascular diseases, obesity.

Methodological works

[†] for co-first authorship, and * for co-corresponding authorship

- F. Wang, M. Chen, J. Li, Y. Yu, **Z. Guo***, M. Zou* (2025), m6Astorm: Stoichiometry-preserving and stochasticity-aware identification of m6A from direct RNA sequencing. *Briefings in Bioinformatics*, 27 (2), bbag132.
- L. Zhang, **Z. Guo**, Z. Qin, Z. Wu (2024), m6A peak calling accounting for varying sequencing bias across regions and samples. **2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)**, 659-662.
- L. Chen[†], **Z. Guo[†]**, T. Deng, H. Wu (2024), scCTS: identifying the cell type specific marker genes from population-level single-cell RNA-seq. **Genome Biology**, 25 (1), 269.
- **Z. Guo[†]**, D. Duan[†], W. Tang, J. Zhu, W. S. Bush, L. Zhang, X. Zhu, F. Jin, H. Feng (2024), magpie: a power evaluation method for differential RNA methylation analysis in N6-methyladenosine sequencing. **PLOS Computational Biology.**, 20 (2), e1011875.
- D. Duan, W. Tang, R. Shu, **Z. Guo***, H. Feng* (2023) Evaluation of Differential Epitranscriptome Analysis Methods. **Briefings in Bioinformatics**, 24 (3), bbad139.
- **Z. Guo**, A. M. Shafik, P. Jin, H. Wu (2022) *Differential RNA Methylation Analysis for MeRIP-seq Data under General Experimental Design*. **Bioinformatics**, 38(20): 4705-4712.
- **Z. Guo**, A. M. Shafik, P. Jin, Z. Wu, H. Wu (2021) *Detecting m6A methylation regions from Methylated RNA Immunoprecipitation Sequencing*. **Bioinformatics**, 37(18): 2818–2824. <https://doi.org/10.1093/bioinformatics/btab181>
- Z. Li, **Z. Guo**, Y. Cheng, P. Jin, H. Wu. Robust partial reference-free cell composition estimation from tissue expression. **Bioinformatics**, (2020), 36(11): 3431-3438. <https://academic.oup.com/bioinformatics/article/36/11/3431/5804977>
- **Z. Guo[†]**, Y. Cui[†], X. Shi, J. A. Birchler, I. Albizua, S. L. Sherman, Z. S. Qin*, T. Ji* (2020) *An empirical bayesian approach for testing gene expression fold change and its application in detecting global dosage effects*. **NAR Genomics and Bioinformatics**, 2(3):lqaa072. <https://academic.oup.com/nargab/article/2/3/lqaa072/5908375>
- **Z. Guo**, S. Zhang (2020) *Sparse deep nonnegative matrix factorization*. **Big Data Mining and Analytics**, 3(1):13-28. <https://ieeexplore.ieee.org/document/8935092>

Collaborative works

- Y. Pan, **Z. Guo**, K. Kathleen, Y. Yang, G. Khalil, R. Kelly, B. Switzer (2025+) *Simulation Study for Improving Efficacy Estimates for New HIV Treatments. To be submitted.*
- M. C. Runnstrom, P. A. Lamothe, ..., **Z. Guo**, ..., F. E. Lee. (2024) *Patients taking benralizumab, dupilumab, or mepolizumab have lower postvaccination SARS-CoV-2 immunity*. **Journal of Allergy and Clinical Immunology**. 154(2):435-446. <https://www.sciencedirect.com/science/article/pii/S0091674924004202>

- M. C. Runnstrom¹, A. Morrison-Porter, M. Ravindran, H. Quehl, R. Ramonell, M. Woodruff, M. Kuruvilla, F. Anam, R. Swerlick, C. Swenson, C. Polito, W. Neveu, R. Patel, C. Kim, I. Hentenaar, **Z. Guo**, H. Wu, N. S. Haddad, J. L. Daiss, I. Sanz, F. E. Lee* (2022), *Reduced SARS-CoV-2 Vaccine Response in Patients Treated with Biologic Therapies for Asthma*. **Journal of Allergy and Clinical Immunology**, 149(2):AB319. [https://www.jacionline.org/article/S0091-6749\(21\)01844-3/fulltext](https://www.jacionline.org/article/S0091-6749(21)01844-3/fulltext)
- A. M. Shafik, F. Zhang, **Z. Guo**, Q. Dai, K. Pajdzik, Y. Li, Y. Kang, B. Yao, H. Wu, C. He, E. G. Allen, R. Duan, P. Jin* (2021) *N6-methyladenosine dynamics in neurodevelopment and aging, and its potential role in Alzheimer's disease*. **Genome Biology**, 22(1):1-19. <https://link.springer.com/article/10.1186/s13059-020-02249-z>

Funding Support

- NSFC Young, 12401650 (PI: Zhenxing Guo) 2025-2027
 Statistical Models and applications for the cross-sectional and longitudinal analysis of differential m6A methylation and gene expression
 Total cost: RMB 300,000.
 Role: PI
 Status: Active
- NSFC Special, 82441027 (PI: Yao Lu) 2025-2027
 Establishment and clinical validation of a rehabilitation treatment strategy for nasopharyngeal carcinoma based on treatment response (Joint Application B)
 Total cost: RMB 1,500,000
 Role: Co-Investigator
 Status: Active

Book Chapters

- **Z. Guo**, A. Shafik, P. Jin, Z. Wu, H. Wu (2023) Analyzing mRNA epigenetic sequencing data with TRESS. In: H. Oliveira, P. (eds) Computational Epigenomics and Epitranscriptomics. Methods in Molecular Biology, vol 2624. Humana, New York, NY. https://doi.org/10.1007/978-1-0716-2962-8_12.

Software Developed

- **magpie** This package aims to perform power analysis for the MeRIP-seq study. It calculates FDR, FDC, power, and precision under various study design parameters. Available on Bioconductor: <https://bioconductor.org/packages/magpie>.
- **TRESS**: Toolbox for mRNA Epigenetics Sequencing analysis. An R/Bioconductor package for analysis of RNA methylation sequencing data. Available at <https://bioconductor.org/packages/devel/bioc/html/TRESS.html>.
 — Software based on publication Guo 2021 **Bioinformatics**; Guo 2022 **Bioinformatics**.

- **Foldseq**: Fold change detection from RNA-seq data. An R package for testing gene expression fold change. Available on github at <https://github.com/cuiyingbeicheng/Foldseq>.
— Software based on publication Guo 2020 **NAR Genomics and Bioinformatics**

Presentations / Invited talks / lectures at other academic institutes

- scCTS: identifying the cell type specific marker genes from population-level single-cell RNA-seq, ICSA China 2024, Wuhan, China. June 2024
- Benchmark of N6-methyladenosine differential analysis methods and a Power Assessment Tool for MeRIP-seq. CCBSB 2023, Qingdao, China. October 2023
- Benchmark of N6-methyladenosine differential analysis methods and a Power Assessment Tool for MeRIP-seq. CSBMB 2023, Tai'an, China. September 2023
- Evaluation of Differential Epitranscriptome Analysis Methods. ICSA China 2023, Chengdu, China. July 2023
- Statistical method and tools for RNA methylation sequencing analysis. Department of Population and Quantitative Health Sciences, School of Medicine, Case Western Reserve University. May 2022
- Statistical method and tools for RNA methylation sequencing analysis. School of Data Science, Chinese University of Hong Kong, Shenzhen. April 2022
- Guo, Z. Differential m⁶A Analysis for MeRIP-seq Data under General Experimental Design. Poster Presentation at Georgia Statistics Day, Atlanta, GA Oct 2021
- Guo Z., Shafik A. M., Jin P., Wu Z., Wu H. (2021) Detecting m6A methylation regions from Methylated RNA Immunoprecipitation Sequencing. JSM 2021, Virtual Aug 2021
- Including Historical Data in Clinical Trial Analysis. Summer school at Shandong University. July 2021

Teaching Experience

At SDS, CUHK-SZ, as the instructor of

- STA3002/STA4102 Generalized Linear Models: Spring 2023, Spring 2024: Spring 2025
- STA2002H Honors Probability and Statistics II, Spring 2025
- STA2002 Probability and Statistics II, Summer 2024
- STA2004 Mathematical statistics Instructor. Spring 2023, Spring 2024.

At Emory, as the teaching assistant of

- Fall 2020: BIOS 731 Advanced statistical computing.
- Spring 2020: BIOS 545 R programming.

- Fall 2019: BIOS 555 High-throughput data analysis using R/Bioconductor.
- Spring 2019: BIOS 591P Lab “Biostatistics Methods II”.
- Fall 2018: BIOS 500/003: Statistical Methods I Group C.

Mentoring experience

At CUHK-SZ

- Ms. Fangyi Peng, PhD student of School of Data Science at CUHKSZ, co-supervised with Prof. Jin Liu
- Ms. Xu Guo, Master student of School of Medicine at CUHKSZ
- Ms. Yang Yu, Master student of School of Medicine at CUHKSZ

At Emory

- PhD student mentor for 1st year theory and 2nd year method qualifying exam, 2020.
- PhD student mentor for 2nd year theory qualifying exam, 2021.

Service Activities

Committee service

- July. 2025 - present: Member, Epi&Biostats MPhil/PhD Programme Admission Committee, School of Medicine and School of Data Science
- Aug. 2022 - present: Member, DS PhD Admission committee, School of Data Science
- Nov. 2022 - present: Member, MSc in BIM Admission committee, School of Medicine and School of Data Science
- June 2023 - present: Member, Seminar committee, School of Data Science
- Aug. 2023 - present: Chair of Library & Web Committee, School of Data Science
- Academic Advisor group - 2024: AA for 21 SDS undergraduate students
- Academic Advisor group – 2023: AA for 33 SDS undergraduate students

Conference program

- Organizer and Chair: ICSA 2023, Topic-contributed Session “Technical advances in analyzing single cell RNA-seq and spatial transcriptomic data.” Chengdu. June 2023
- Chair: ICSA China 2023, Topic-contributed Session “Addressing Computational Challenges in Analyzing High-throughput Genomic and Epigenetics Data”. Chengdu. June 2023.
- Program Committee: IEEE International Conference on Bioinformatics and Biomedicine (IEEE BIBM). 2023 & 2024 & 2025
- Program Committee: IEEE International Conference on Medical Artificial Intelligence (IEEE MedAI). 2023 & 2024

Editorial Activities

Journal Reviewer

- Annals of Applied Statistics, Briefings in Bioinformatics, Genome Biology, Bioinformatics Advances, Current Bioinformatics, Scientific Reports, Statistical Methods in Medical Research

Honors and Award

- Presidential Young Fellow, CUHK-SZ, 2025.
- Honorable Mention, Emory Biostatistics senior PhD student presentation competition, 2022.
- Excellent Poster Award of Georgia Statistics Day, 2021.
- Michael H. Kutner Doctoral Student Award, Emory University, 2021-2022.
- JSM Distinguished Student Paper Award, 2021.

Professional Membership

- American Statistical Association (ASA)
- The International Biometric Society — Eastern North American Region (ENAR)
- International Chinese Statistical Association (ICSA)

Computational Skills

- Programming Languages: R, SAS, Matlab, Python
- Operating Systems: Windows, Unix